Now you need to plan out your loop. There are several readily available sequences that you can follow. Depending on how many things you wish to cool with your loop, you'll have to throw in additional components. The basic setup is to have a reservoir which is fed by a pump. The reservoir leads to a radiator which then leads to the first cooling block and then to all subsequent blocks. Once all blocks are interconnected, the output leads to the pump inlet. Here's a handy diagram for the same.

Based on your requirements, you'll have to come up with your loop sequence and then get all the components. What components you ask? These:

- **CPU Block** - This is the chunk of metal that sits on your CPU.
- **GPU Block** - Same as the CPU block but for your graphics card. The GPU block is slightly larger because it also covers the VRM circuitry which heats up quite a lot.
- **Pump** - The heart of your cooling system that keeps the coolant flowing in it.
- **Reservoir** - This is where you'd have reserve coolant, should some dry up or should you top up a little too much sometime.
- **Radiator** - The coolant once it has cooled the entire system lands up in the radiator where the heat is radiated out and the coolant gets cooled.
- **Fans** - While a radiator can do the job on its own if it is big enough, having cooling fans speeds up the job. If the radiator fins are dense, then you'll have to get fans which are capable of pushing air through the dense fins. Basically, what you're looking for are fans with high static pressure.

While these components are interconnected with tubing, you need something to keep the tubing firmly attached to the components. That's where fittings come in. Based on what kind of tubing are being used, you'll need the appropriate fitting. For soft tubing, you can use barbed or compression fittings. For hard-tubing, you have push-in fittings. Remember the ID and OD we talked about previously? Keep that in mind while getting the fittings, especially for hard tubing since getting the wrong size means your tubes won't stay put at all. Alternatively, you can use compression fittings for hard tubing as well, except this isn't the same as the one used for flexible tubing.

Once you have all the components, fittings and the tubings. You'll need the appropriate tools. Especially, with hard tubing. To bend hard tubing you need a hot air gun and then templates to bend your tube with. Templates

Silicone insert to help with bends

using a 45 and 90 degree angle are the most common. To get the desired angle, you would heat a section of the tube and then place it on the template to get the proper angle. However, there is a problem that all tubes face when you bend them and that's a collapse. To prevent a tube from collapsing, you can use a temporary filler in this case or as they're more commonly known - silicone inserts.

With flexible tubing, you have a similar issue of unwanted collapses. For such cases, you get plastic coils which can be used as a support structure. Just wind some around whichever section is most prone to bending and you're set.

The next step is to build the loop based on your plan. Once everything is set, you have to begin the filling. Be extra careful with these steps. Usually, this is done via the reservoir which has a priming inlet. Simply keep pouring until the reservoir is filled to the brim. Then wait a while and let the coolant course through the tubings, once it comes to a halt, commence the filling again. There will be a point where you cannot pour any more coolant, this is when you insert a tube onto the priming inlet and raise its opening above the case. This gives you more headroom to get more coolant in. Repeat until you cannot fill in more coolant. Once you've hit the limit. Close the priming inlet and run the loop. There are going to be numerous air bubbles stuck in different components of your loop and letting it run for a while will get all of them to collect in the reservoir. Once that happens, you'll have more space to add coolant. Rinse repeat until you can add no more. At this point, you should have successfully finished your loop.

The next step is to run the loop and observe carefully. Sometimes leaks don't appear during the filling stage but the extra pressure during running can make them appear clearly. If the loop runs fine for a good 24-hours without any issue, you will have completed your first custom liquid cooling setup. Congratulations!

CASE MODDING

The most difficult and the most creative of all mods is the case mods. The possibilities are diverse and the scene is very vibrant. There are even multiple national and international case modding competitions to bring the best in the world together. Speaking of the best, we even managed to talk to one of the top modders in the country, Rakesh Sharma, who was more than enthusiastic to share his latest work with us and you.

Dubbed the Nighthawk build, it's inspired by its namesake, the Lockheed F-117 Nighthawk fighter aircraft. As of writing this, the build hasn't finished but it's nearly done. And instead of explaining all the steps, we're simply going to let his build log speak for itself.